

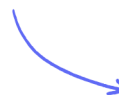
IGCSE · Cambridge (CIE) · Maths

 25 mins  4 questions

Modelling: App Downloads

Modelling

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Total Marks

/25

- 1 (a)** This task looks at the release of a new game app, in particular finding models for the number of downloads in the short term and in the long term.

An app company, called Concept, plans to release a new game app which will be available to download for free from a popular app platform.

The app company will work together with an advertising agency who will use their advertising power to increase the number of downloads.

The app company proposes an exponential-growth model to predict the total number of downloads, y , measured t **days** after the app is released, given by

$$y = 5 \times 2^{kt}$$

where k is a positive value.

They initially use the value $k = 0.5$

Find the total number of downloads predicted by the model 16 days after the app is released.

(1 mark)

- (b)** Show that the model **first exceeds** 1 billion total downloads 56 days after releasing the app.

Give any supporting working in standard form to 3 significant figures.

(2 marks)

- (c)** Decide whether or not the model is suitable for predicting the total number of downloads 1 year after the app is released.

Explain your answer.

(1 mark)

- (d)** The advertising agency tells the app company that a more realistic value of k for them to use in their model is

$$k = 0.079376$$

The advertising agency also warns the app company that exponential growth is only expected in the first 180 days after releasing a new app.

The first 180 days is referred to as the exponential stage.

With the new value of k , show that there will be approximately 100 000 total downloads by the end of the exponential stage.

(1 mark)

- 2 (a)** An app company, called Concept, plans to release a new game app which will be available to download for free from a popular app platform.

The first 180 days after releasing the app is referred to as the exponential stage.

The app company believes it can reach 100 000 total downloads by the end of the exponential stage.

After the exponential stage, the rate at which the app is downloaded starts to slow down over the next 10 years, called the deceleration stage.

The app company decide to use a **logarithmic** model to predict the total number of downloads, Y , in **millions**, after n **years** into the deceleration stage, given by

$$Y = a + 0.88 \log(n + 1)$$

The value of $n = 0$ indicates the start of the deceleration stage (and the end of the exponential stage).

Show that if $a = 0.1$, then the number of downloads at the start of the deceleration stage matches the number of downloads at the end of the exponential stage.

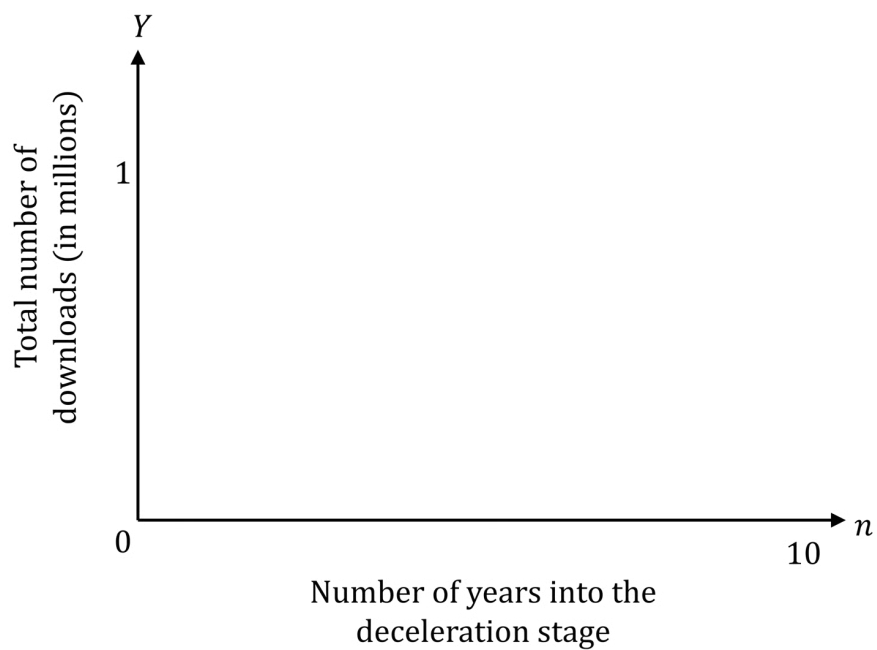
(2 marks)

- (b)** Sketch a graph of the logarithmic model

$$Y = 0.1 + 0.88 \log(n + 1)$$

on the axes provided.

Label any axes intercepts clearly.



(2 marks)

- (c)** Find, in millions, the total number of downloads predicted after 2 years into the deceleration stage.

Give your answer correct to 3 significant figures.

(1 mark)

- 3 (a)** An app company, called Vision, has designed a game app which they plan to release and have available for free download from a popular app platform.

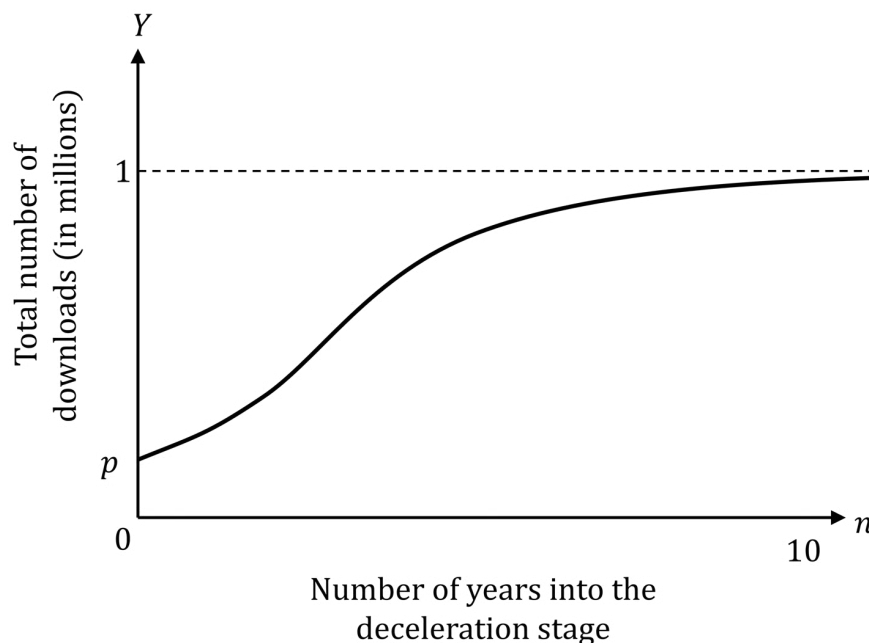
The initial 180 days after releasing the app is referred to as the exponential stage.

After the exponential stage, the rate at which the app is downloaded starts to slow down over the next 10 years, called the deceleration stage.

Vision is using a **logistic** model to predict the total number of downloads, Y , in millions, after n years into the deceleration stage, given by

$$Y = \frac{1}{1 + 9 \times 2^{-n}}$$

A sketch of the logistic model is given below, where the line $Y = 1$ is a horizontal asymptote.



Find the value of p .

(1 mark)

- (b)** The company wants to know how long it would take for their predicted total number of downloads to reach half a million, $Y = \frac{1}{2}$.

Use algebra to show that, if $Y = \frac{1}{2}$, then n satisfies the equation

$$2^n = 9$$

(3 marks)

(c) Solve the equation

$$2^n = 9$$

to find out how many years (into the deceleration stage) it takes for the total number of downloads to reach half a million.

Show your working clearly.

Give your answer correct to 3 significant figures.

(2 marks)

(d) Explain why this logistic model is not a good model for the company to use if their goal is to achieve over 1 million total downloads.

(1 mark)

- 4 (a)** Two app companies, the first called Concept and the second called Vision, are releasing similar game apps to be available for free download from a popular app platform.

For both companies, the initial 180 days after releasing the app is referred to as the exponential stage.

After the exponential stage, the rate at which the app is downloaded starts to slow down over the next 10 years, called the deceleration stage.

Concept uses the following logarithmic model for its app, to predict the total number of downloads, Y , in millions

$$Y = a + 0.88 \log(n + 1)$$

Vision uses the following logistic model for its app, to predict the total number of downloads, Y , in millions

$$Y = \frac{1}{1 + 9 \times 2^{-n}}$$

In both cases, n represents the number of years into the deceleration stage. The value of $n = 0$ indicates the start of the deceleration stage (and the end of the exponential stage).

The first company, Concept, are worried that the predicted total number of downloads from the second company, Vision, is higher than theirs.

Show that, after 7 years into the deceleration stage, the predictions from each model **support** this concern.

(2 marks)

- (b)** The company Concept decides to investigate the two models further, by plotting them on the same axes.

Find, as an inequality in n , the range of years over which Vision predicts a higher total number of downloads than Concept.

Give your answer correct to 2 significant figures.

(3 marks)

- (c) Explain clearly how the inequality in part (b) could be used to reassure Concept that they do not need to worry about the results in part (a).

(1 mark)

- (d) Concept decide that they do not want to have any period of time in which Vision has a higher predicted number of downloads compared to them.

They decide to vary the constant 0.88 in their logarithmic model by adapting the model as follows,

$$Y = 0.1 + \left(0.8 + \frac{c}{50}\right) \log(n+1)$$

where c can take any positive integer value.

When $c = 4$, the original model is obtained.

By investigating integer values of c that are greater than 4, find the smallest integer value of c that ensures the predicted total number of downloads by Concept is **always** greater than that of Vision.

Explain your answer.

(2 marks)